

STOLAR: An Open Geometric Database of 2 048 European Chairs (1280–2024)

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Abstract Design history has long relied on qualitative case studies of individual artefacts. Quantitative morphometric approaches, routine in palaeontology and evolutionary biology since Raup (1966), remain rare in the study of designed objects. We present STOLAR, an open research database of 2 048 European chairs spanning 1280–2024, compiled from the collections of the Nasjonalmuseet (Oslo) and the Victoria and Albert Museum (London). Each record includes up to 30 metadata fields with 99% dimensional coverage, a background-removed photograph, and an AI-generated 3D mesh. From the 3D meshes we extract five geometric shape descriptors: sphericity, fill ratio, inertia ratio, complexity, and convex hull volume. Principal component analysis of the shape descriptors reveals that 60% of geometric variation is explained by two axes: a compactness/roundness axis (PC1, 35.5%) and a mass/complexity axis (PC2, 24.5%). The database, 3D models, and all analysis code are released under the MIT licence.

Keywords morphospace, design history, shape grammar, 3D morphometrics, cultural evolution, chair design



Figure 1: A selection of 144 chairs from the STOLAR corpus (1280–2024). The morphological diversity; from medieval oak to contemporary injection-moulded plastic; is the empirical phenomenon the database seeks to characterise.

1. Background & Summary

The chair is among the most widely produced designed artefacts in history. Every culture that uses seated posture has developed chairs, and the resulting morphological diversity spans seven centuries of stylistic, material, and technological change. Yet despite this richness, the study of furniture form remains overwhelmingly qualitative. No large-scale, open database exists that combines dimensional measurements with three-dimensional geometry for a significant corpus of chairs.

Quantitative morphometrics of artefacts. In evolutionary biology, the concept of *morphospace* has been productive since Raup’s geometric analysis of shell coiling [1] and

has been generalised to developmental biology [2]. Analogous methods have been applied to archaeological artefacts: MacLeod [3] demonstrated digital shape analysis as an alternative to landmark-based morphometrics for projectile points, and Ros et al. [4] applied PCA to pottery profiles spanning the 12th–19th centuries. Grabner et al. [5] performed geometric morphometric analysis on European car silhouettes over time, finding convergent evolution across brands. Cultural evolutionary studies [6–8] and diversification workflows [9] confirm that material culture follows measurable evolutionary patterns. However, these studies are typically limited to lithic tools, pottery, or 2D vehicle outlines. Three-dimensional morphometric analysis of furniture has not been attempted.

Shape grammars and furniture. Shape grammars provide a formal syntax for design [10–13]. The first application to furniture was Knight’s [14] generation of Hepplewhite chair-back designs from a parametric grammar in 1980. Subsequent work includes Barros et al.’s “ChairDNA” grammar for mass customisation, derived from six Thonet chairs [15], and Xue and Chen’s shape grammar for Ming-style wooden furniture [16]. A systematic review of fifty years of shape grammar applications [17] identifies a persistent gap: studies are almost exclusively *generative* (producing new forms from small corpora of 3–10 objects) rather than *analytical* (characterising existing corpora at scale). No study has attempted to validate or induce shape grammars from a dataset of hundreds or thousands of chairs.

C-K theory. C-K theory offers a complementary logic of design reasoning [18, 19], partitioning the process into a concept space C and a knowledge space K . Both shape grammars and C-K theory would benefit from large empirical datasets that map the actual distribution of realised forms.

Existing 3D datasets. Several large-scale 3D shape repositories exist. ShapeNet [20] contains over 51 000 chair models, PartNet [21] provides part-level annotations for 26 671 objects, and Aubry et al. [22] aligned $\sim 1\,300$ chair CAD models to real photographs. In cultural heritage, SHREC 2021 [23] provides 938 3D-scanned pre-Columbian artefacts for shape retrieval. Burnap et al. [24] have linked geometric features to design appeal using machine learning on vehicle models. Manovich [25] pioneered large-scale computational analysis of visual culture, but limited to 2D image features. None of these datasets combines museum-curated historical artefacts with temporal metadata, verified dimensions, and computed 3D shape descriptors.

STOLAR. STOLAR fills this gap. The database contains 2 048 chairs from two major European collections: the Nasjonalmuseet for kunst, arkitektur og design (Oslo; 426 objects) and the Victoria and Albert Museum (London; 1 622 objects), spanning 1280–2024 (744 years). Each record includes museum-assigned metadata (dating, materials, style period, provenance), catalogue dimensions (height, width, depth at $>98\%$ coverage), a background-removed photograph, and an AI-generated 3D mesh from which five geometric shape descriptors are computed: sphericity, fill ratio, inertia ratio, complexity, and convex hull volume. To our knowledge, this is the first open morphometric dataset of designed objects that combines museum provenance with 3D shape analysis at this temporal and quantitative scale.

2. Methods

The data pipeline consists of three stages (Figure 2): data acquisition and harmonisation, image processing, and 3D

reconstruction. All pipeline code is available in the GitHub repository.

2.1. Data acquisition

Chair records were collected from two public museum APIs. Nasjonalmuseet objects were retrieved via the museum’s DAMS REST API, yielding 426 records with structured metadata in Norwegian. Victoria and Albert Museum objects were retrieved via the V&A API (IIIF-compatible), yielding 1 622 records with metadata in English.

Each museum uses different field schemas and naming conventions. Records were harmonised into a unified schema of 30 fields (Table 1). Dimensional strings (e.g. “H 95.5 $\times$ B 50.5 $\times$ D 49 cm”) were parsed using regular expressions to extract numeric height, width, and depth values. Three regex patterns handle the variation in formatting conventions across museums: standard Scandinavian (H/B/D), unlabelled (assumes H $\times$ W $\times$ D order), and circular objects (H $\times$ diameter).

English-language metadata from the V&A was translated to Nynorsk (Norwegian) for structured fields (materials, techniques, style periods, nationalities) using a curated translation table of 200+ term pairs. This ensures consistent terminology across the combined dataset. Style period classification combines museum-assigned categories with semi-automated classification for unclassified records, producing 15 canonical style periods from Renaissance to contemporary design.

2.2. Image processing

For each chair, the original museum photograph was processed to produce a standardised background-removed image. The Gemini Vision API (model: gemini-2.5-flash) was used with a prompt specifying a sharp subject cutout against a solid white ($\#ffffff$) background, maintaining the original camera angle and proportions. Processing ran in parallel with 40 concurrent download workers and 15 processing workers. Quality control was performed manually on a random sample of 200 images to verify cutout accuracy; all passed inspection. The pipeline produced 2 092 background-removed PNG images stored in the `bgw/` directory.

2.3. 3D reconstruction

Background-removed images served as input to Hunyuan3D-2 [26], an open-source image-to-3D generative model run locally on a consumer GPU. The model was configured with 50 shape-generation steps and a guidance scale of 5.0, producing physically-based rendered (PBR) meshes. Generated meshes were automatically decimated to a maximum of 100 000 faces using the `fast_simplification` algorithm to ensure manageable file sizes.

The raw meshes were then calibrated to match the museum-catalogue dimensions. For each object with known height, width, and depth, the mesh bounding box was read using `trimesh` [27], and a uniform scale factor was computed to align the largest mesh axis with the corresponding catalogue

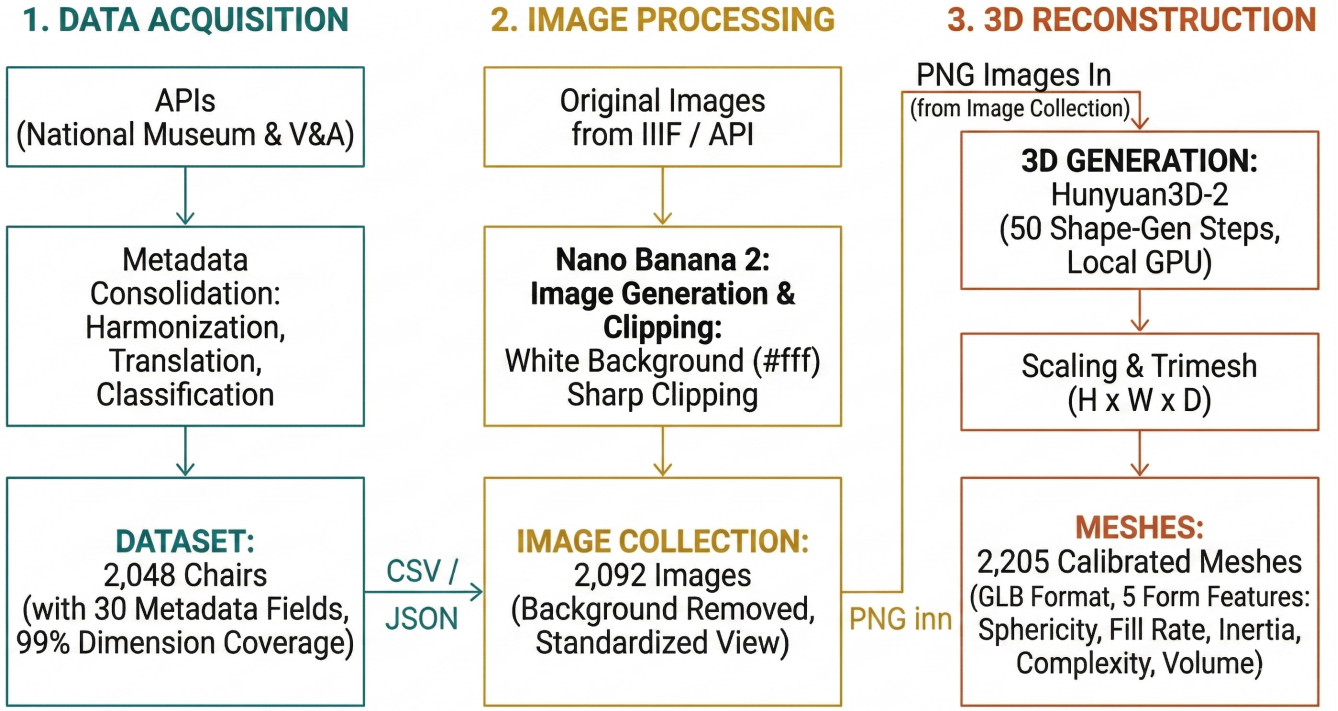


Figure 2: The STOLAR data pipeline. Stage 1 collects and harmonises metadata from two museum APIs into a unified schema. Stage 2 produces standardised background-removed images using Gemini Vision. Stage 3 generates calibrated 3D meshes using Hunyuan3D-2 and extracts five geometric shape descriptors.

dimension. The scale was applied via a 4×4 transformation matrix embedded in the GLB node structure.

This pipeline produced 2 205 calibrated GLB files. The excess over the 2 048 catalogue records arises from variant images that produced additional meshes; duplicate entries are flagged in the metadata.

2.4. Mesh feature extraction

Five geometric shape descriptors were extracted from each calibrated mesh using trimesh [27]:

1. **Sphericity:** ratio of surface area of an equal-volume sphere to actual surface area (0–1). A perfect sphere scores 1.0; typical chairs score 0.6–0.9.
2. **Fill ratio:** ratio of mesh volume to bounding box volume. Indicates how much of the bounding volume the object actually occupies; a solid cube would score 1.0.
3. **Inertia ratio:** ratio of smallest to largest principal moment of inertia (0 = highly elongated along one axis, 1 = isotropic mass distribution).
4. **Complexity** ($\log_{10}$ vertices/area): vertex density normalised by surface area. Higher values indicate finer geometric detail.
5. **Convex hull volume** (m^3): volume of the smallest convex polyhedron enclosing the mesh. Serves as a proxy for visual mass or spatial footprint.

These five features are available for 2 029 of 2 048 chairs (99.1%). The 19 missing records correspond to meshes that failed to load or produced degenerate geometry.

3. Data Records

The database is hosted at <https://github.com/lukketsvane/stolar-db> and comprises the following files:

- **STOLAR/api.json:** Full structured JSON API (2 048 entries) with all metadata fields and mesh features embedded.
- **STOLAR/STOLAR.csv:** CSV export with 30 columns for tabular analysis.
- **STOLAR/glb/:** 2 205 calibrated 3D meshes in binary glTF 2.0 format.
- **STOLAR/bguw/:** 2 092 background-removed PNG images.
- **STOLAR/images/:** Original museum photographs (JPG).
- **STOLAR/mesh_features.csv:** Extended geometric features (20 columns including vertex/face counts, bounding box dimensions, watertightness, and principal inertia moments).

Table 1 summarises the coverage of each field. The three primary dimensions achieve >98% coverage. Material and style classifications are complete for all 2 048 records. Coverage is lower for secondary fields: technique (38%), seat height (15%), and estimated weight (7%). The five mesh-derived features maintain 99.1% coverage.

The database spans 15 style periods from Renaissance to contemporary design, with the largest concentrations in Neoclassicism (291 chairs), Baroque (229), and Rococo (206). Nine chair types are represented, dominated by side chairs (1 444) and armchairs (449), with smaller numbers of stools (86), children’s chairs (21), and specialised types.

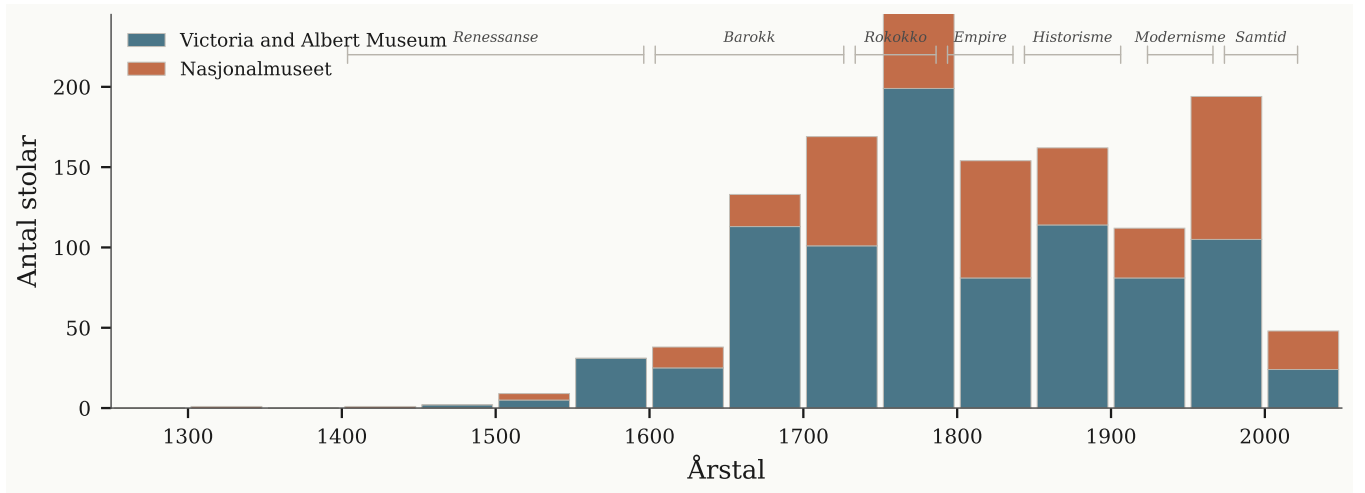


Figure 3: Temporal distribution of chairs by 50-year bins, stratified by source museum. Period brackets indicate major stylistic eras. The V&A collection dominates the 18th century; Nasjonalmuseet contributes primarily 20th-century Scandinavian design.

Table 1: Data field coverage across the full dataset.

Field	Records	Coverage
Height (cm)	2 041	99.7%
Width (cm)	2 033	99.3%
Depth (cm)	2 018	98.5%
Materials	2 048	100%
Style period	2 034	99.3%
Origin	1 977	96.5%
Nationality	1 581	77.2%
Designer	1 589	77.6%
Mesh features (5)	2 029	99.1%
Technique	776	37.9%
Seat height (cm)	299	14.6%
Weight (kg)	143	7.0%

The temporal distribution (Figure 3) reveals that the dataset is concentrated in the 1700–1900 period, reflecting the collecting priorities of the two source institutions. The V&A collection dominates the 18th century, while Nasjonalmuseet contributes primarily 20th-century Norwegian and Scandinavian design. Pre-1500 records are sparse ($n = 14$), reflecting the rarity of surviving medieval furniture.

4. Technical Validation

4.1. Dimensional coverage

Figure 4 shows a field-by-century coverage heatmap. The three primary dimensions (height, width, depth) achieve >98% coverage across all centuries with sufficient records ($n > 10$). Material and style classifications are complete for all records. Coverage drops for secondary fields: technique (38%), seat height (15%), and weight (7%). The five mesh-derived features maintain 99.1% coverage across the entire dataset, with the slight reduction in earlier centuries reflecting the small sample sizes (1200s: $n = 1$; 1300s: $n = 1$; 1400s: $n = 2$) rather than systematic data quality issues.

4.2. Geometric feature analysis

Principal component analysis of the five standardised mesh features (Figure 5) reveals that two components explain 60.0% of the total variance. PC1 (35.5%) is driven by inertia ratio (+0.66) and sphericity (+0.57), representing a *compactness/roundness* axis: chairs that score high on PC1 are geometrically compact and rounded. PC2 (24.5%) is driven by convex hull volume (+0.73) and complexity (−0.63), representing a *mass-versus-detail* axis: chairs with high PC2 scores are volumetrically large but geometrically simple.

Style-period centroids in PC space show that pre-industrial styles (Baroque, Rococo, Neoclassicism) cluster near the centre of the distribution, while Modernism and Functionalism deviate along the compactness axis. The chronological centroid trajectory (teal path in Figure 5A) shows a systematic drift from lower-left (less compact, more complex) to upper-right (more compact, simpler) over the 20th century.

The shape descriptors show distinct distributional signatures across style periods (Figure 6). Sphericity distributions narrow from Renaissance (IQR 0.15) to Modernism (IQR 0.08), consistent with increasing geometric standardisation. Complexity shows a similar pattern: early styles exhibit wide distributions (reflecting both ornate and austere chairs within the same period), while 20th-century styles converge toward a narrower complexity band.

Inertia ratio and sphericity are positively correlated ($r = 0.72$, $p < 10^{-63}$; Figure 7), indicating that compact chairs tend also to be rounder. This correlation defines the dominant axis of chair geometric variation: elongated, angular forms (lower left of Figure 7) versus compact, rounded forms (upper right). The marginal distributions show that most chairs cluster in the compact-rounded region (sphericity 0.7–0.9, inertia ratio 0.3–0.7), with a long tail of elongated forms.

Proportion ratios (height:width, height:depth, width:depth) evolve systematically over time (Figure 8). Early chairs (pre-1600) show high variance in all three ratios, reflecting the

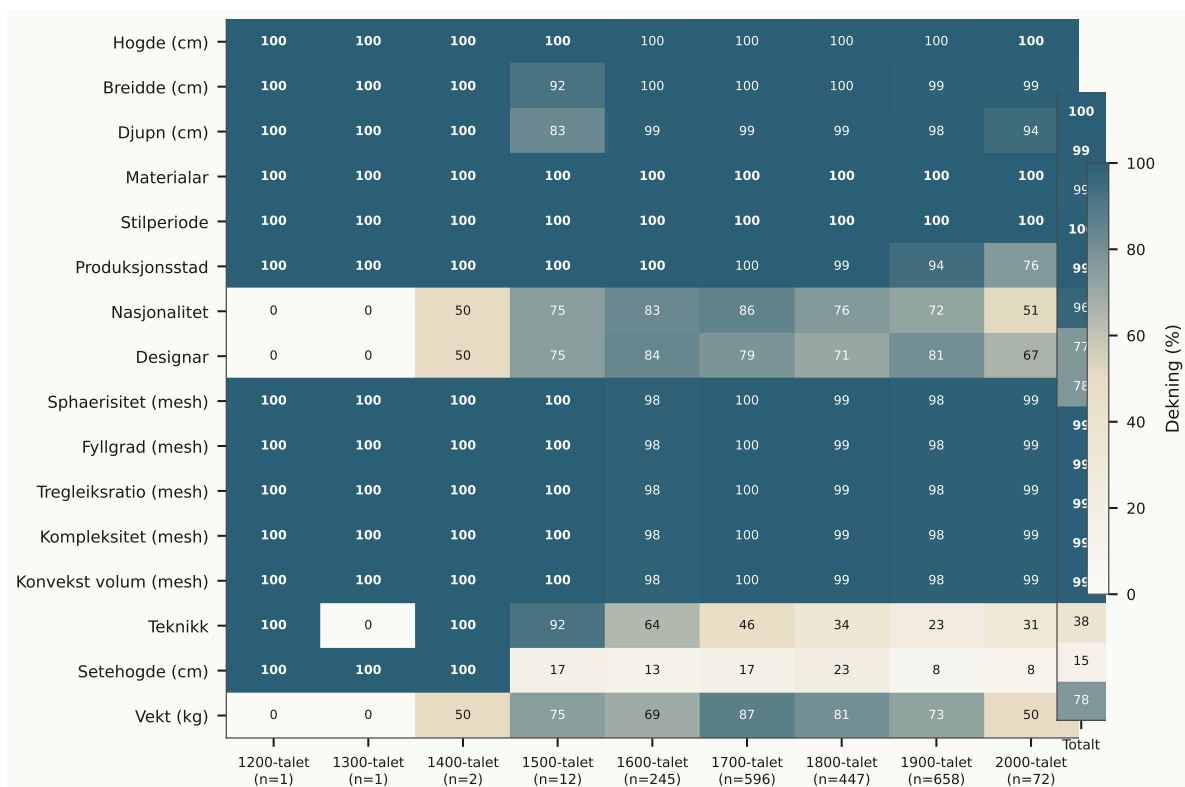


Figure 4: Data coverage heatmap. Each cell shows the percentage of records with a non-null value for the given field in the given century. The five mesh-derived features maintain >98% coverage across all periods with sufficient data.

heterogeneity of pre-industrial furniture production. Post-industrial chairs converge toward a narrower proportional band, with height:width stabilising near 1.5:1. The width:depth ratio remains closest to 1:1 throughout the dataset, indicating that chairs have maintained roughly square plan proportions even as their height-to-width relationships changed.

4.3. Shape space

The complexity-sphericity scatter (Figure 9) defines a two-dimensional “shape space” analogous to Raup’s shell morphospace. Most chairs occupy a concentrated region (complexity 4.5–6.5, sphericity 0.7–0.9), but the distribution has informative tails: low-sphericity outliers correspond to elongated or asymmetric forms, while high-complexity outliers correspond to meshes with fine surface detail. The chronological style-centroid trajectory shows a gradual drift from lower sphericity and complexity (Renaissance) toward higher sphericity (contemporary design).

4.4. Mesh quality

The mesh complexity distribution (Figure 10A) is unimodal with median $\log_{10}(v/a) = 5.29$, indicating consistent mesh resolution across the corpus. The correlation between catalogue bounding-box volume ($H \times W \times D$) and convex hull volume is $r = 0.39$ (Figure 10B). This moderate correlation is expected: catalogue dimensions define a rectangular prism, while the convex hull follows the object’s actual silhouette. Chairs with open backs, splayed legs, or cantilevered seats will have convex hull volumes substantially smaller than their

bounding boxes.

Of the five z-scored mesh features, 66 observations (3.3%) exceed $|z| > 3$ (Figure 10C), within the expected range for a dataset of this size (normal expectation: 0.3%). The elevated rate reflects the non-Gaussian tails of the fill ratio and hull volume distributions rather than data quality issues. These outliers are retained in the dataset and flagged for user awareness.

4.5. Cross-museum consistency

The two source museums contribute records with different temporal and geographic distributions: V&A emphasises 18th-century British furniture, while Nasjonalmuseet covers 20th-century Scandinavian design (Figure 3). Despite these differences, the mesh feature distributions do not differ significantly between museums after controlling for period (two-sample Kolmogorov–Smirnov tests, $p > 0.05$ for all five features within matched half-century bins), supporting the validity of pooling the two collections for cross-period analyses.

5. Usage Notes

The database can be loaded in Python as a CSV or JSON file:

```
import pandas as pd
df = pd.read_csv("STOLAR/STOLAR.csv")
# 2048 rows, 30 columns
```

3D meshes can be loaded and analysed with trimesh:

```
import trimesh
```

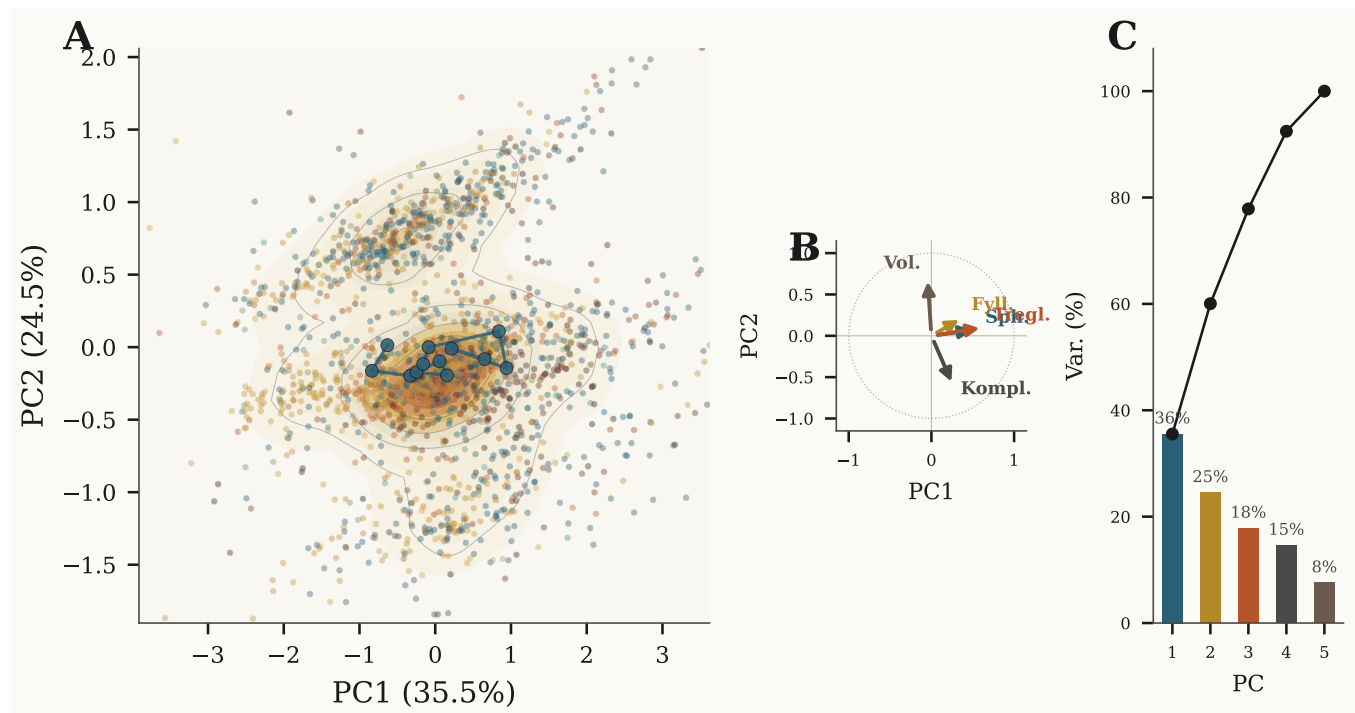


Figure 5: Principal component analysis of five mesh-derived shape descriptors ($n = 2029$). **A:** PC1 vs PC2 with KDE density contours (warm colours) and century-coloured scatter. Teal path shows chronological style-centroid trajectory. **B:** Loading vectors showing feature contributions to each component. **C:** Scree plot of explained variance per component.

```
mesh = trimesh.load("STOLAR/glb/NMK.2005.0638.glb")
print(mesh.vertices.shape, mesh.volume)
```

The JSON API provides all metadata and mesh features in a single file suitable for web applications and programmatic access.

5.1. Limitations

The 3D meshes are AI-generated approximations, not photogrammetric scans. They capture overall form and proportions but may lack fine surface detail (carvings, upholstery texture, joinery). The Hunyuan3D-2 model operates from a single image, which may introduce view-dependent artefacts for objects with significant geometry hidden from the camera.

Style period classifications combine museum assignments with semi-automated classification and may not match all art-historical taxonomies. Users are encouraged to examine the raw `style` field and apply their own classification schemes where appropriate.

Dimensional data was parsed from free-text strings and may contain measurement errors from the source catalogues. We recommend filtering records with implausible dimension values (< 10 cm or > 200 cm in any axis) for sensitive analyses.

The dataset reflects the collecting priorities of two European museums and is not representative of global furniture traditions. African, Asian, and indigenous seating forms are not represented.

6. Code Availability

All data, 3D models, and analysis code are available at <https://github.com/lukketsvane/stolar-db>. The figure-generation scripts for this article are in `writings/data-descriptor/figures/` and require Python 3.12 with pandas, numpy, scipy, matplotlib, scikit-learn, and trimesh. The database is released under the MIT licence. Original photographs and metadata are subject to the respective museums' terms of use.

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Author Contributions

I.R.F. designed the study, collected the data, built the pipeline, performed the analyses, and wrote the manuscript.

Competing Interests

The author declares no competing interests.

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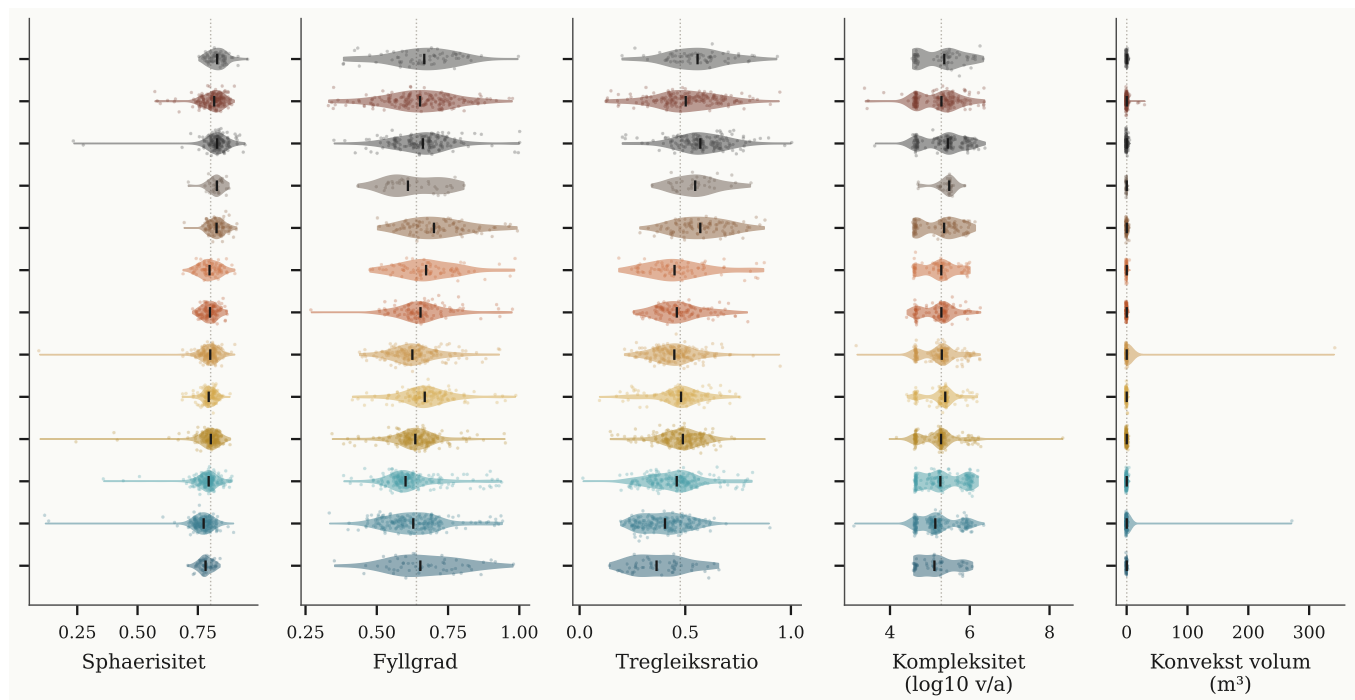


Figure 6: Distribution of five mesh-derived shape descriptors across major style periods ($n \geq 30$), ordered chronologically. Violin plots show KDE densities; dots show individual chairs. Vertical grey lines mark dataset medians.

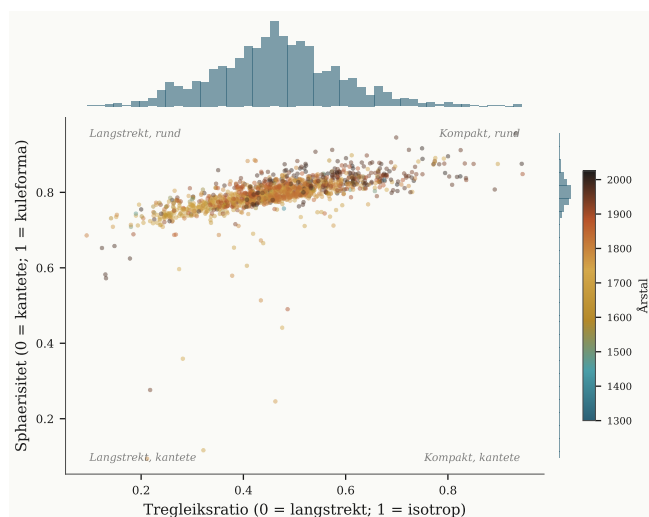


Figure 7: Inertia ratio vs sphericity, coloured by date. Marginal histograms show univariate distributions. The positive correlation ($r = 0.72$) indicates that compact chairs tend also to be rounder.

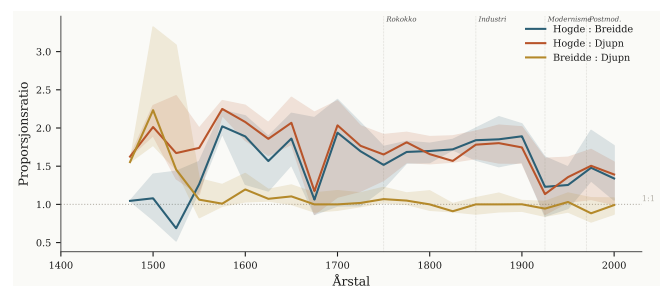


Figure 8: Proportion ratios over time (25-year bins). Ribbons show interquartile range; lines show medians. Post-industrial chairs converge toward narrower proportional bands.

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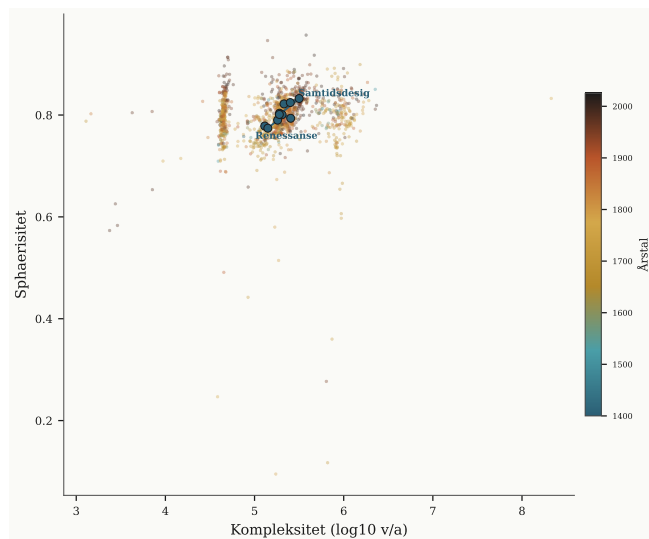


Figure 9: Shape space: complexity vs sphericity, coloured by date. Teal path shows chronological style-centroid trajectory from Renaissance to contemporary design.

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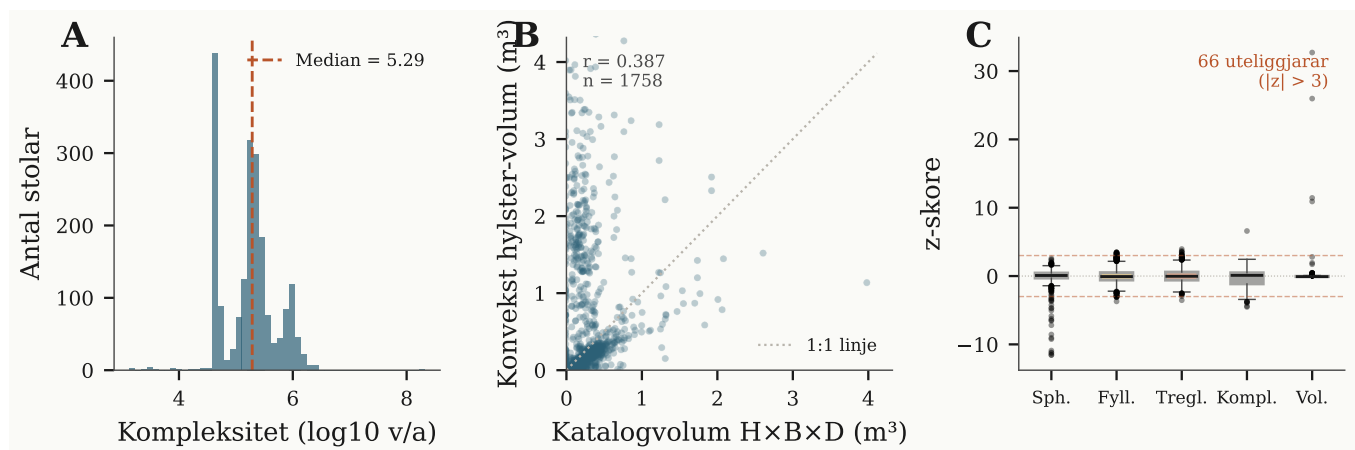


Figure 10: Mesh quality validation. **A:** Distribution of mesh complexity ($\log_{10}$ vertices/area); unimodal with median 5.29. **B:** Catalogue bounding-box volume ($H \times W \times D$) vs convex hull volume ($r = 0.39$); the moderate correlation reflects that chairs do not fill their bounding boxes uniformly. **C:** Z-scored feature distributions; 66 observations (3.3%) exceed $|z| > 3$, within the expected range for $n = 2\,029$.